

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - HIGH (Pseudo-code Writing)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO1: *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 10

Total Marks: 50

ANNEXURE A

PSEUDO-CODE WRITING

Code of Conduct:

I, A K Selva Prabhu (192421224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A.K. Selva

1) Student Management System :

CLASS student

PRIVATE

name

rollNo

marks1, marks2, marks3

PUBLIC

METHOD acceptDetails()

INPUT name

INPUT rollNo

INPUT marks1, marks2, marks3

END METHOD

METHOD CalculateAverage()

average = (marks1 + marks2 + marks3) / 3

RETURN average

END METHOD

METHOD displayResult()

avg = CalculateAverage()

IF avg >= 50 THEN

PRINT "PASS"

ELSE

PRINT "Fail"

END IF

END METHOD

END CLASS

START

CREATE STUDENT S

S.acceptDetails

S.displayResult()

STOP

T. Calculate Avg()

Library Book Management

PRIVATE

title

author

ISBN

is Available = TRUE

PUBLIC

METHOD IssueBook()

IF isAvailable = TRUE THEN

is Available = FALSE

PRINT "Book Issue"

ELSE

PRINT "Book Not Available"

ENDIF

END METHOD

METHOD returnBook()

is Available = TRUE

PRINT "Book Returned"

END Method.

METHOD displayDetails()

Print title

Print author

Print ISBN

Print is Available

END Method

END class

START

CREATE Book B1

CREATE Book B2

CREATE Book B3

B1.Issue Book()

B1.return Book()

B1.display Details()

STOP

3) Employee Salary Calculation

CLASS Employee

PRIVATE

EmpId

name

basicPay

designation

PUBLIC

METHOD Calculate Gross()

HRA = basicPay * 0.20

DA = basicPay * 0.10

gross = basicPay + HRA + DA

RETURN gross

END Method.

METHOD Calculate Net()

gross = Calculate Gross()

PF = gross * 0.05

Tax = gross * 0.10

net = gross - PF - Tax

Return net

END Method.

METHOD display Payslip()

PRINT empId

PRINT name

PRINT designation

PRINT Calculate net()

END Method.

Shape Area Calculator (Polymorphism)

CLASS Shape()

METHOD calculateArea()

END METHOD

END CLASS

CLASS Circle EXTENDS Shape

METHOD calculateArea()

$area = 3.14 * radius * radius$

PRINT area

END METHOD

END CLASS

CLASS Rectangle EXTENDS Shape

METHOD calculateArea()

$area = length * breadth$

PRINT area

END METHOD

END CLASS

CLASS Triangle EXTENDS Shape

METHOD calculateArea()

$area = 0.5 * base * height$

PRINT area

END Method

END class

START

CREATE Circle C

CREATE Rectangle R

CREATE Triangle T

C. calculateArea()

R. calculateArea()

T. calculateArea()

5). STOP.
Vehicle Registration System

class Vehicle.

regno.

owner

fuelType

METHOD displayDetails()

PRINT regno.

PRINT owner

PRINT fuelType

END Method.

END CLASS

CLASS TwoWheeler EXTENDS Vehicle

Engine Capacity

METHOD displayDetails()

PRINT regno

PRINT owner

PRINT fuelType

PRINT engine Capacity

END Method.

END CLASS

CLASS FourWheeler EXTENDS Vehicle

seating Capacity

AC status

Method displayDetails()

PRINT regno

PRINT owner

PRINT fuelType

PRINT seatingCapacity

PRINT AC status

END METHOD

END CLASS.

5.

Bank Account Hierarchy. 7)

CLASS Bank Account

acc no.

holder

balance

METHOD withdraw (amount)

END Method.

END CLASS

CLASS Savings Account EXTENDS
Bank Account

METHOD withdraw (amount)

if balance > amount THEN

balance = balance - amount.

Else

PRINT "Insufficient Balance"

ENDIF

END Method

END class

CLASS Current Account EXTENDS
Bank Account.

METHOD withdraw (amount)

balance = balance - amount

PRINT "Overdraft Allowed"

END Method

END class

Hospital Patient Record System.

Class Patient.

PRIVATE

diagnosis

prescription

Method generate Bill()

END Method

END CLASS

CLASS INPATIENT EXTENDS PATIENT

daily charge

days

METHODS generate Bill()

bill = daily charge * days.

PRINT bill

END METHOD

END CLASS

CLASS out Patient EXTENDS
PATIENT

Consultation Fee

Method generate Bill()

PRINT consultation Fee

END METHOD

END CLASS

8. E-Commerce Product Catalog.

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CLASS Product
METHOD apply Discount
END Class

CLASS Electronics EXTENDS Product
METHOD apply Discount()
  PRINT "10% Discount"
END Method.
END Class

CLASS Clothing EXTENDS Product
METHOD apply Discount()
  PRINT "20% Discount"
END Method.
END CLASS

CLASS Grocery EXTENDS Product.
METHOD apply Discount()
  PRINT "5% Discount"
END Method.
END CLASS

```

9) University Staff Hierarchy

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Class Staff
  Method Calculate Allowance()
END Class

Class TeachingStaff EXTENDS Staff
  Method Cal. Allowance()
    Print "Teaching Allowance"
END CLASS

Class Professor Ex. TeachingStaff
  Method Cal. Allowance()
    Print "Professor Allowance"
END Class

Class NonTeachingStaff Ex. Staff
  Method Cal. Allowance()
    Print "Non-Teaching Allowance"
END Class

START
Create Professor P
Create NonTeachingStaff N
P. Calculate Allowance()
N. Calculate Allowance()

```


b) Smart Home Device Controller.

Class SmartDevice

Method executeCommand(cmd)

END class

CLASS SmartLight Extends SmartDevice

Method executeCommand(cmd)

PRINT "Light: " + cmd

END class

CLASS SmartLock Extends SmartDevice

PRIVATE PIN

Method executeCommand(cmd)

Print "Lock: " + cmd

End class

Start

Create SmartLight L

Create SmartLock S

L executeCommand("ON")

S executeCommand("Lock")

STOP.